

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. BPE-601: Topics in Modern Physics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **five** of the following: [5 × 4 = 20]

- (a) State and explain the Cosmological Principle.
- (b) What are degenerate stars? Explain briefly with an example.
- (c) Draw neatly the Hertzsprung-Russell (H-R) diagram and explain it.
- (d) Write down the Lorentz transformations for energy and momentum when the boost is along the x -axis.
- (e) What are Stokes and anti-Stokes lines in Raman scattering?
- (f) What are gauge transformations? Show that the electric and magnetic fields remain invariant under gauge transformations.

Question 2.

- (a) Derive an expression for the kinetic energy of the recoiled electron in Compton scattering. [8]
- (b) What is the physical meaning of Compton wavelength? Justify your answer. [4.5]

OR

- (a) Derive the Hamiltonian for a charged particle moving under the influence of an electromagnetic field. [8]
- (b) Explain briefly the working of a p-n junction diode as a radiation detector. [4.5]

Question 3.

- (a) Discuss the Big Bang nucleosynthesis. [6]
- (b) Explain the differences between Big Bang nucleosynthesis and stellar nucleosynthesis. [6.5]

OR

- (a) What are source-free Maxwell's equations? Write these equations in four-vector notation. [8]

- (b) Derive the relationship $m = \frac{m_0}{\sqrt{1-v^2/c^2}}$, where m_0 is the rest mass of the particle, m is the relativistic mass, and v is the velocity of the particle. [4.5]
-

Question 4.

- (a) Derive a condition for the hydrostatic equilibrium of a mass of gas contracting under gravity. [4.5]
- (b) Explain the process of generation of energy by the Sun. Discuss its evolution and ultimate fate (i.e., its death). [8]

OR

- (a) State and explain the significance of Hubble's law. [4]
- (b) What is cosmic microwave background? Explain its origin. [4]
- (c) What are the various structures in the visible universe? What does it mean by large-scale smoothness of the universe? [4.5]